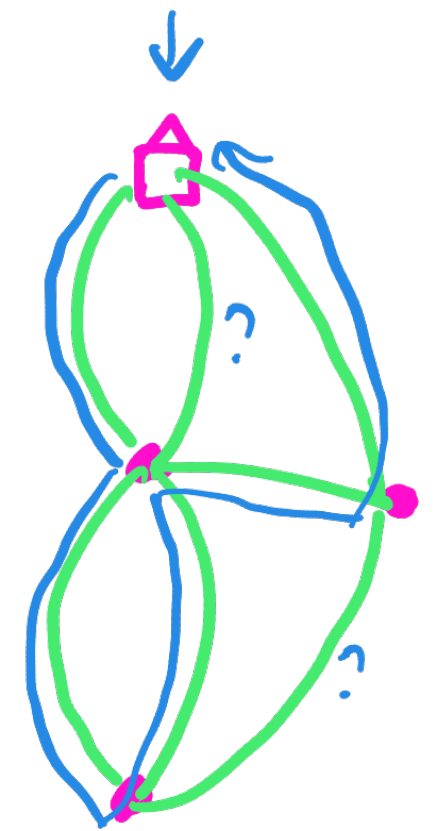
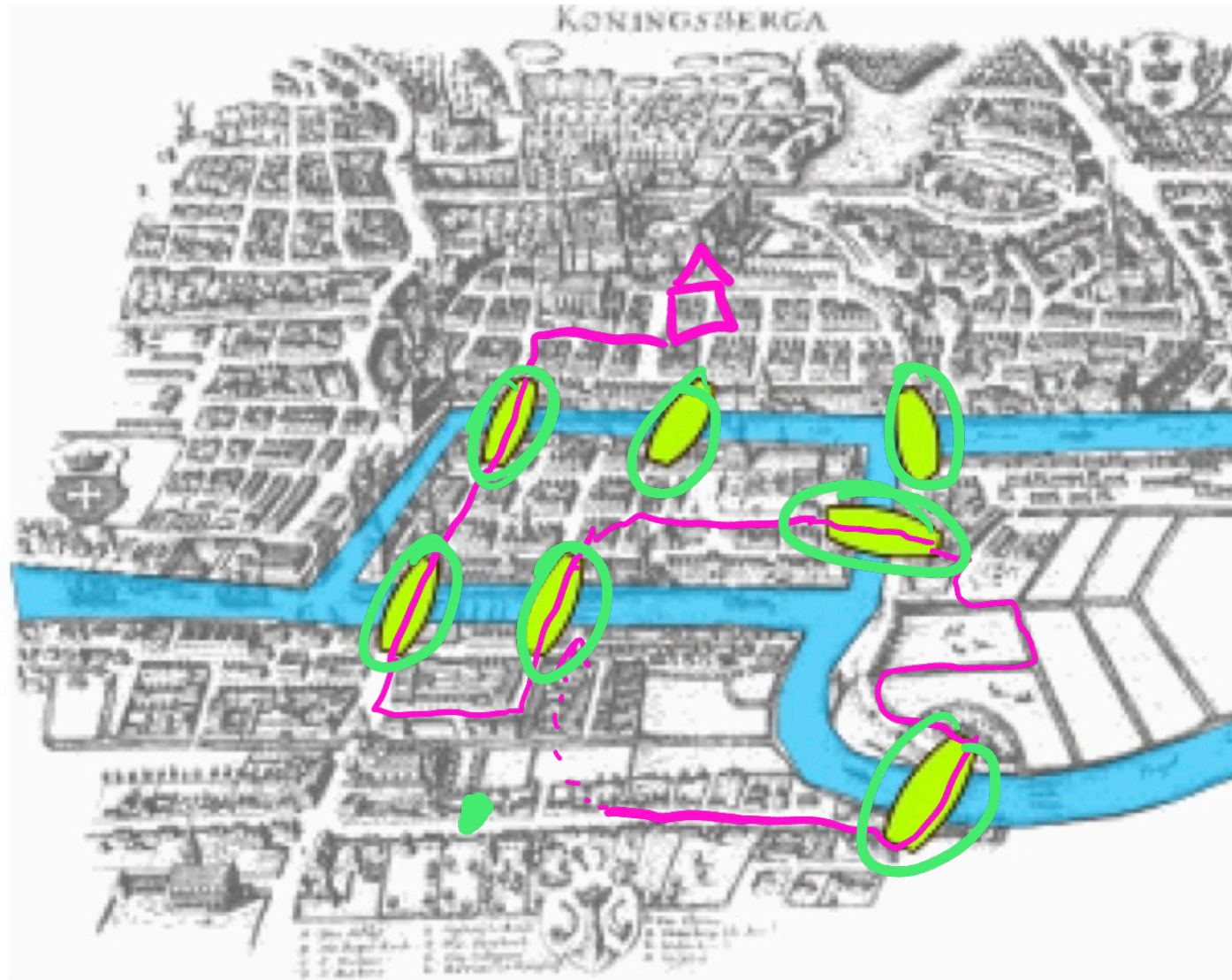


Lecture Notes: Graph Theory—Euler Paths and Circuits

Math F113X Math and Society

Konigsberg and Leonhard Euler [“oiler”]



Is there a way to travel on every edge exactly once and return home?

Terminology

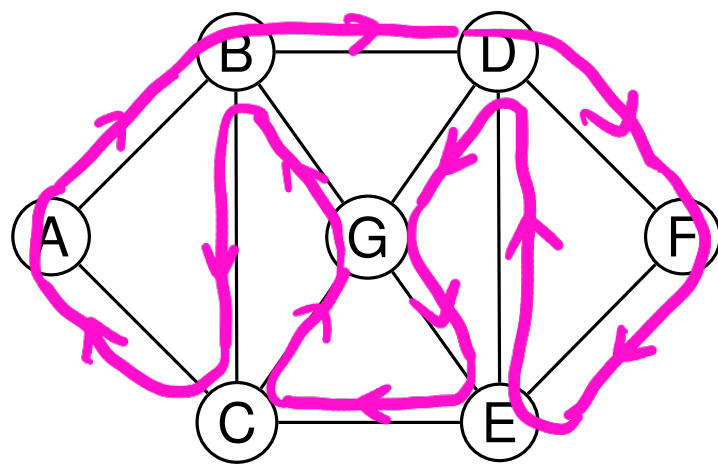
Euler Path

A path on a graph that uses every edge exactly once

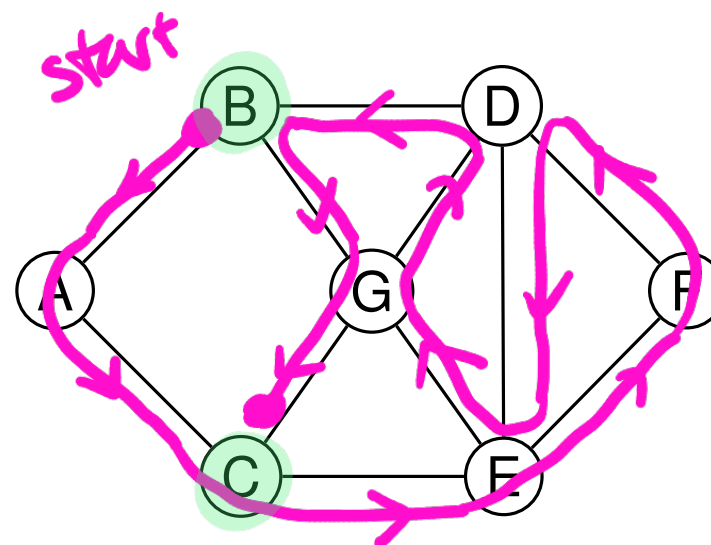
Euler Circuit

An Euler path that is a circuit — it starts and ends at the same vertex

Euler Paths and Circuits



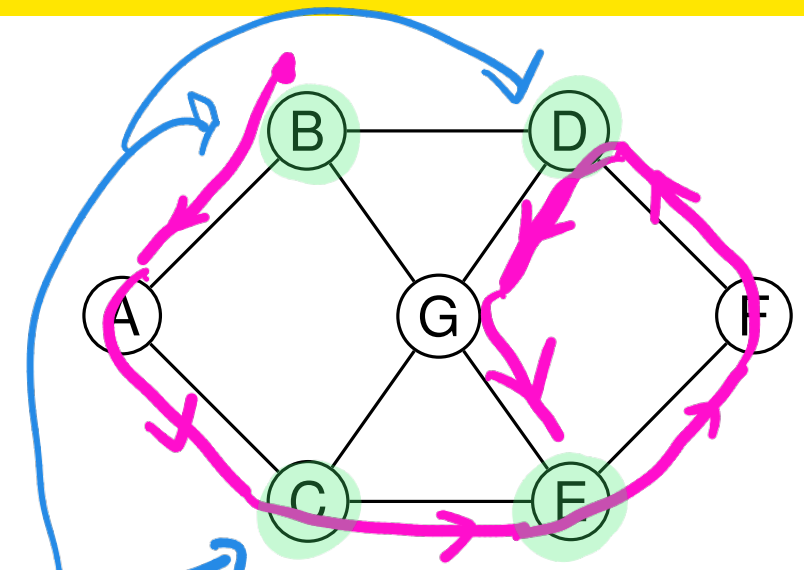
Euler Circuit!



stop
Euler Path

Not a circuit
started at B and
ended at C

No Euler circuit



odd
degree

Neither Euler
Path nor
a circuit

Highlight odd-degree
vertices!

A path uses up 2 edges every time it goes through a vertex!

Applications

Snow plowing

← plows need to go along every road in a neighborhood

edges = roads

vertices = intersections

Mail Delivery

← hit every house in a neighborhood